Project: Retail Sales Data Warehouse and Analytics Engineering with dbt

Bussiness Scenario

You are a Data Engineer/Analytics Engineer for an online retail company. The company stores customer, product, order and order-item information in a PostgreSQL operational database. Management wants to understand sales performance, identify its most profitable product categories and analyse monthly sales trends.

Your task is to design an analytical data warehouse and use dbt to transform the operational data into reliable, tested datasets for business reporting.

Find a dataset that matches your business case and highlight the reasoning behind the choice of dataset.

Project Requirements

1. Design the Data Warehouse

Identify the sales business process and declare the grain of your fact table. Design a star schema containing a `fact_sales` table and appropriate dimension tables, such as `dim_customers`, `dim_products` and `dim_date` or something equivalent.

Create a dimensional model diagram showing the relationships between the fact and dimension tables. Explain the measures and descriptive attributes stored in each table.

2. Implement the Warehouse Using dbt

Create a dbt project connected to your PostgreSQL database. Define the operational tables as dbt sources and create staging models to standardise column names, data types and basic data-quality issues.

Create an intermediate model if needed to combine order and order-item information. Then build the final fact and dimension models based on your Week 9 design.

3. Test and Document the Data

Implement dbt tests to check that primary or business keys are unique, required fields are not null, and fact-table foreign keys match valid dimension records.

Create at least two additional business-rule tests. For example, test for invalid order quantities or sales amounts that violate your stated business rules. Document your models and important columns so another analyst can understand and use them.

4. Analyse the Business Data

Draft business questions and use SQL to answer the business questions using your completed warehouse:

Your queries must use the fact and dimension models rather than querying the raw operational tables directly.

Expected Deliverables

Submit a GitHub repository containing:

- Dimensional model diagram and declared fact-table grain.
- SQL/dbt project files, including source definitions, staging models and fact/dimension models.
- dbt tests and model documentation.
- SQL queries answering the business questions.
- A README explaining the business problem, warehouse design, project structure, setup instructions and how to run the models and tests.
- Evidence of successful model execution and test results.

A Powerpoint Presenation for the demo. Remember to assume the role that I have described above and come prepared to the demo with some ownership. This is what we talk about almost every week. Thank you.